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Consensus recommendations for the use of noninvasive melanoma detection techniques based on results of an international Delphi process

To the Editor: Noninvasive techniques for facilitating melanoma detection include total body photography (TBP), serial digital dermoscopic imaging (SDDI), and reflectance confocal microscopy (RCM). Recommendations for implementing these technologies in clinical practice are lacking; therefore, a committee of dermato-oncologists performed an online Delphi process to obtain consensus opinion regarding appropriate use of these techniques.

The principal investigators (R.A.W. and J.G.K.) selected 13 committee members based on their dermato-oncology experience. The principle investigators developed 24 clinical scenarios that were sent to committee members who evaluated each scenario using a 5-point Likert scale that ranged from strongly agree to strongly disagree. Committee members were asked to provide rationale for their position on each recommendation. At the end of the first round, all committee members were asked to propose scenarios for evaluation during the next round. Recommendations were considered to represent consensus of the committee if 75% or more of the committee members reported that they agreed/strongly agreed or disagreed/strongly disagreed with the statement (Table I). At the end of the second round, 11 consensus statements were generated (Table II).

Patient need for TBP is based on a combination of mole phenotype and baseline risk for developing melanoma.[1] Other considerations that support using TBP include presence of significant photodamage, patient history of many biopsies, and patient anxiety. Although children do not typically require TBP given the volatility of their nevi and generally low risk of developing melanoma, patients with genetic syndromes or large congenital nevi may benefit from TBP. Once baseline images are obtained, new images do not need to be retaken until the patients' skin lesions have significantly evolved, resulting in difficulty comparing their dermatologic examination to their photographs.

Pigmented lesions with obviously malignant dermoscopic features should be biopsied in toto. SDDI is used for lesions with equivocal dermoscopic features and can be used alone or in conjunction with TBP for monitoring new or changing lesions.[1] The interval that lesions undergoing monitoring with SDDI should be reassessed is based on prospective and retrospective studies and is dependent on the lesion phenotype (eg, macule/patch vs papule/plaque).[2,3] Some members dissented from recommendation #1, preferring biopsy or RCM for "ugly duckling" pigmented lesions with equivocal dermoscopic features.[4]

RCM is recommended when biopsy is indicated but patient factors (eg, cosmetically sensitive areas, sites with delayed healing, or patient hesitation about a biopsy) make noninvasive testing preferable, because prospective data support that RCM can detect melanoma while also sparing patients unnecessary biopsies.[5] For melanoma in situ, RCM allows patients to undergo only 1 definitive therapeutic procedure; whereas, for invasive melanoma, biopsy is still necessary to determine Breslow depth. Repeat RCM is not typically performed after negative imaging but can be considered for borderline or changing lesions.

These techniques can help facilitate earlier diagnosis of melanoma and can spare patients unnecessary biopsies. These consensus recommendations aim to help clinicians appropriately

Table I. List of all discussed scenarios

Total body photography—Round 1
Total body photography is recommended for evaluation of patients with a personal history of malignant melanoma
Total body photography is recommended for evaluation of patients with either greater than 5 clinically atypical nevi or a personal history of biopsy proven dysplastic nevus
Total body photography is recommended for evaluation of patients with a first-degree relative (eg, parent, sibling, child) with a history of malignant melanoma
Total body photography is recommended for evaluation of patients with either known genetic mutations that confer risk of malignant melanoma or a syndrome that confers risk of malignant melanoma
Total body photography is recommended for evaluation of adults with greater than 100 nevi who are otherwise at average risk for malignant melanoma
Total body photography is recommended for evaluation of children with greater than 100 nevi who are otherwise at average risk for malignant melanoma
Total body photography is recommended for evaluation of children with clinically atypical nevi AND a first-degree relative (e.g. parent, sibling, child) with a history of malignant melanoma
Total body photography is recommended for evaluation of children with a history of a Spitz nevus
In the above scenarios, total body photography is preferable to serial digital photography when available. Serial digital photography is recommended in the above scenarios when total body photography is not available
When utilizing total body photography new images should be obtained at every total body skin examination
When utilizing total body photography new images should be obtained once a year
When utilizing total body photography new images should be obtained if the patient has had many skin biopsies or is developing new melanocytic lesions
When available, total body photography should be performed in conjunction with serial digital dermoscopy imaging for evaluation of clinically atypical nevi
Total body photography—Round 2
Total body photography is recommended for patients with familial atypical multiple mole melanoma syndrome (FAMMM Syndrome, aka dysplastic nevus syndrome)
Total body photography is recommended in adults with more than 50 nevi that have 1 or more of the following: a personal history of multiple malignant melanomas, a personal history of an amelanotic melanoma, multiple pink nevi, multiple clinically atypical nevi, a genetic syndrome that predisposes to the development of malignant melanoma.
Total body photography is NOT recommended for all patients with >50 nevi in the absence of atypical nevi, changing nevi, pink nevi, or significant risk factors for melanoma
Total body photography is NOT recommended for all patients with a history of a solitary melanoma in the absence of >50 nevi, multiple clinically atypical nevi, pink nevi, or multiple changing nevi.
Total body photography is generally NOT recommended in children given the volatility of their nevi and their low risk of developing a malignant melanoma.
When utilizing total body photography it is recommended that new images should be obtained once previous images are not easily comparable to the patients current appearance
When performing total body photography, the use of serial digital dermoscopy imaging and/or serial digital photography is also recommended for evaluation of clinically atypical nevi
Serial digital dermoscopic imaging—Round 1
Serial digital dermoscopic imaging is recommended in patients with less than 5 clinically atypical nevi who are otherwise at average risk for development of MM
Initial repeat digital dermoscopic imaging is recommended 3 months after initial images are obtained.
Serial digital dermoscopic imaging is recommended in patients with a medium to giant size congenital melanocytic nevus
Serial digital dermoscopic imaging is recommended in patients with a large or growing lentigo-like lesion
In the above scenarios, serial digital dermoscopic imaging is preferable to serial digital photography when available. Serial digital photography is recommended in the above scenarios when serial digital dermoscopic imaging is not available
Serial digital dermoscopic imaging is not recommended for evaluation of lesions that are highly concerning for malignant melanoma
Initial repeat digital dermoscopic follow-up images are recommended 6 months after the initial images are obtained.
In patients with a personal history of for malignant melanoma, a first-degree relative with for malignant melanoma, greater than 5 clinically atypical nevi, or greater than 100 nevi serial digital dermoscopic imaging is not recommended for use as a solo modality for digitally evaluating nevi; however, in these scenarios it is recommended as an adjunct to total body photography

Continued

Table I. Cont'd

Serial digital dermoscopic imaging—Round 2
Serial digital dermoscopy is recommended for monitoring ''ugly duckling'' nevi with equivocal dermoscopic features
Except when evaluating lentigo-like lesions, lesions undergoing digital dermoscopic imaging should be re-evaluated at a 3-month interval
Serial digital dermoscopy imaging is recommended in patients with a large or growing lentigo-like lesion that lack diagnostic dermoscopic features with a plan to re-evaluate at a 6-month interval
Reflectance confocal microscopy—Round 1
Reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal pigmented lesions in cosmetically sensitive areas or in areas prone to poor wound healing
Reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal pigmented lesions with dermoscopic signs of regression
Reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal amelanotic lesions
Reflective confocal microscopy is recommended for evaluation of dermoscopically equivocal pigmented lesions on children or patients that are reluctant to undergo biopsy
Reflectance confocal microscopy is recommended for evaluation of suspected lentigo maligna
Reflectance confocal microscopy is recommended for patients with clinically atypical nevi with reassuring dermoscopic features that are the cause of significant psychosocial distress
Reflectance confocal microscopy is recommended for all pigmented lesions
Reflectance confocal microscopy is recommended for changing melanocytic lesions that are not definitely malignant
Reflectance confocal microscopy is recommended to help distinguish an actinic keratosis from a squamous cell carcinoma
Reflectance confocal microscopy is recommended to help rule out a desmoplastic melanoma
Reflectance confocal microscopy is recommended to determine margins of a neoplasm
Reflectance confocal microscopy—Round 2
When available, reflectance confocal microscopy is recommended for monitoring response of lentigo maligna undergoing treatment with imiquimod
Reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal pink and/or amelanotic lesions with a plan for re-evaluation in 3 months for lesions with negative reflectance confocal microscopy
Reflectance confocal microscopy is not recommended as a tool for providing reassurance to patients with nevi with dermoscopically reassuring features that are the cause of significant psychosocial distress.

Table II. Consensus recommendations

Consensus recommendations	Result
Total body photography	
Recommendation #1: Total body photography is recommended for patients with familial atypical multiple mole melanoma syndrome (FAMMM Syndrome, aka dysplastic nevus syndrome)	Strongly agree: 91% Agree: 0% Neither agree nor disagree: 9% Disagree: 0% Strongly disagree: 0%
Recommendation #2: Total body photography is recommended in adults with more than 50 nevi that have 1 or more of the following: (1) a personal history of multiple cutaneous melanomas; (2) a personal history of an amelanotic melanoma; multiple pink nevi, multiple clinically atypical nevi, and/or; (3) a genetic syndrome that predisposes to the development of cutaneous melanoma.	Strongly agree: 64% Agree: 36% Neither agree nor disagree: 0% Disagree: 0% Strongly disagree: 0%
Recommendation #3: Total body photography is generally NOT recommended in children given the volatility of their nevi and their low risk of developing a malignant melanoma.	Strongly agree: 64% Agree: 36% Neither agree nor disagree: 0% Disagree: 0% Strongly disagree: 0%

Continued

Table II. Cont'd

Consensus recommendations	Result
Recommendation #4: When utilizing total body photography, it is recommended that new images should be obtained once previous images are not easily comparable to the patient's current appearance	Strongly agree: 55% Agree: 36% Neither agree nor disagree: 9% Disagree: 0% Strongly disagree: 0%
Recommendation #5: When performing total body photography, the use of serial digital dermoscopy imaging and/or serial digital photography is also recommended for evaluation of clinically atypical nevi	Strongly agree: 45% Agree: 36% Neither agree nor disagree: 18% Disagree: 0% Strongly disagree: 0%
Serial digital dermoscopic imaging	
Recommendation #1: Serial digital dermoscopic imaging is recommended for monitoring "ugly duckling" nevi with equivocal dermoscopic features	Strongly agree: 55% Agree: 27% Neither agree nor disagree: 0% Disagree: 0% Strongly disagree: 18%
Recommendation #2: Serial digital dermoscopic imaging is recommended in patients with a large or growing lentigo-like lesion that lack diagnostic dermoscopic features with a plan to re-evaluate at a 3- to 6-month interval	Strongly agree: 27% Agree: 55% Neither agree nor disagree: 18% Disagree: 9% Strongly disagree: 0%
Reflectance confocal microscopy	
Recommendation #1: When available, reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal pigmented lesions in cosmetically sensitive areas or in areas prone to poor wound healing	Strongly agree: 80% Agree: 10% Neither agree nor disagree: 0% Disagree: 10% Strongly disagree: 0%
Recommendation #2: When available, reflective confocal microscopy is recommended for evaluation of dermoscopically equivocal pigmented lesions on children or patients that are reluctant to undergo biopsy	Strongly agree: 70% Agree: 10% Neither agree nor disagree: 10% Disagree: 0% Strongly disagree: 10%
Recommendation #3: When available, reflectance confocal microscopy is recommended for evaluation of suspected lentigo maligna with a consideration for reimaging in 6 months should initial imaging be negative for malignancy	Strongly agree: 70% Agree: 10% Neither agree nor disagree: 10% Disagree: 0% Strongly Disagree: 10%
Recommendation #4: When available, reflectance confocal microscopy is recommended for evaluation of dermoscopically equivocal pink and/or amelanotic macular lesions with a plan for reevaluation in 3 months for lesions with negative reflectance confocal microscopy (this may not be suitable for palpable lesions)	Strongly agree: 18% Agree: 64% Neither agree nor disagree: 9% Disagree: 9% Strongly disagree: 0%

implement these techniques into practice. These recommendations do not exhaust all possible indications for these techniques and are not meant to replace physician judgment.

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Drs Waldman, Grant-Kels, Curiel, Curtis, Hu, Kerr, Marghoob, Markowitz, Pellacani, Rao, Scope, and Swetter have no conflicts of interest to declare.

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Prevalence of delta-like protein 3 expression in Merkel cell carcinoma

To the Editor: Merkel cell carcinoma (MCC) is a rare but aggressive cutaneous neuroendocrine neoplasm. The United States has experienced a significant increase in the incidence of MCC.[1] Several causative factors of MCC have been identified, including ultraviolet radiation exposure, Merkel cell polyomavirus (MCPyV), and immunosuppression. Although surgery remains the preferred approach to treatment of resectable primary MCC, studies are currently being performed to investigate the role of (neo)adjuvant therapy. Although immunotherapy has completely changed the landscape of treatment for MCC, objective response rates range from 33% to 68% in studies, leaving room for further improvement.[2]

Delta-like 3 protein (DLL3) acts as an inhibitory ligand of NOTCH receptors and is thought to be involved in neurogenesis during early embryonic development.[3] DLL3 has minimal to no surface expression in normal adult tissues, but it is highly upregulated in small cell lung carcinoma and other tumor types of neuroendocrine origin.[4] Given its lack of expression on normal cells, DLL3 presents a promising target for treatment of various solid tumors with neuroendocrine differentiation. Rovalpituzumab tesirine (Rova-T) is a first-in-class antibody-drug conjugate directed against DLL3.[5]

Here, we present results from the first study, to our knowledge, to evaluate the expression of DLL3 in MCC. Formalin-fixed, paraffin-embedded whole and tissue microarray sections from 56 patients with MCC were obtained and immunostained for DLL3 (SP347). The extent of cytoplasmic or membranous staining at any intensity was scored as those adopted in the phase I clinical trial of Rova-T for small cell lung carcinoma (negative, <1%; low, 1%-49%; high, ≥50%).[5] In addition, MCPyV large T-antigen expression was measured in all tumors with CM2B4 antibody staining.

We found that DLL3 expression was seen in 51 of 56 (91%) from MCC samples (DLL3-high, 33 samples and DLL3-low, 18 samples) (Fig 1). Of the matched primary and metastatic tumors, all samples showed a concordant staining pattern between the primary and metastatic tumor (4 samples). We also looked at the relationship between MCPyV status and DLL3 expression and found that 42 of 56 (75%) MCC tumors were positive for MCPyV. DLL3 expression was seen in all (42 of 42, 100%) MCPyV-positive tumors and 9 of 14 (64%) MCPyV-negative tumors (Table I).

To our knowledge, this is the first study to evaluate DLL3 expression in MCC. Although the